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CK

Time to reach Nash Equilibrium

Calvin Kwong Understanding Nash equilibrium (/learn/game-theory-introduction/module/CkoRM/discussions) · 4 months ago (/learn/game-theory-introduction/discussions/D_OiV44kEeW7Cw7aHtmkKw)

Hi,

introduction/profiles/4a31ac67f4e6f9b0f521890c30895840)

I am just wondering in the few experiments quoted in this video, different countries take different rounds to reach the Nash equilibrium.

Is there any relationship between the "degree of rationality" and the time to reach the Nash equilibrium? For example, if the groups are formed by people of similar degree of rationality, do people who are more rational reach the Nash equilibrium more faster than those that are less rational?

Thanks!

coursera (/)

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Hi Calvin, I'm going to post something irrelevant. Are we the only people taking this course???

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IS

israel mendonca dos santos · 4 months ago (/learn/game-theory-introduction/discussions/D_OiV44kEeW7Cw7aHtmkKw/replies/xAX5y5MvEeWy0Q7ABZMsnQ)

Now you're not.

introduction/profiles/f9c5562964a94042fa82dd4367b34978)

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N

novelalizzy · 4 months ago (/learn/game-theory-introduction/discussions/D_OiV44kEeW7Cw7aHtmkKw/replies/n1NiYJUSEeWKNwpBrKr_Fw)

hi, I just started taking this course some few days ago. Will update soon.

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introduction/profiles/65ccc5025ec17ee0f2381a8f0ebc3e00)

Muhammad Adil · 4 months ago (/learn/game-theory-introduction/discussions/D_OiV44kEeW7Cw7aHtmkKw/replies/zl3srjkwEeW9ThFn3E1-Q)

Game theory is the thing that decides that who is more successful in the game, It means that If, you have a best reply or choice of strategy then you are more successful. so, we can say that It depends.

introduction/profiles/78eed569579c0a2057fde1d8d9bb019)

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SM

Satoshi Matsuzawa Teaching Staff · 4 months ago (/learn/game-theory-introduction/discussions/D_OiV44kEeW7Cw7aHtmkKw/replies/nxkp7ZnNEeWy0Q7ABZMsnQ)

That is a very interesting and profound question. Differences in speed of convergence to the Nash equilibrium are indeed an interesting phenomenon.

introduction/profiles/d4428359e1f955dd4ef1f8bfd90a6b01)

I agree with your opinion that the smartness of people has something to do with the speed of convergence. However, I do not think that the phenomenon is explained by differences in the degree of "rationality." In the lecture, a rational person is defined as a one that chooses a better alternative over worse ones, and this has nothing to do with the reasoning ability of others' decision making. Instead, I think that the phenomena should be associated with the ability to reason about others' action and, moreover, the ability to reason about others' reasoning ability.

I would like to explain my idea with an example. Suppose that I participate in an auction. At the first round, everyone submitted bid at five hundred points. How should I bid in the second round? If I believe that other participants would stick to the first bid, I would bid five hundred and fifty points. If I believe that everyone could reason like me, then I would bid six hundred points. If I believe that everyone would end up bidding six hundred points, then I would bid seven hundred points. In the end, I may bid one thousand points if I believe that others have good reasoning skills.

Thus, I believe that differences in rounds to reach the Nash equilibrium can be explained by differences in ability to reason

others' action and the expectation of others' reasoning ability.

The reasoning I illustrated here is called level-k theory. You may find some materials when you google this word.

Game theory shows that if trial-and-error adjustment ever reaches a stable outcome, it ought to be a Nash equilibrium (otherwise, someone deviates and increase her/his payoff). Game theory, however, as it stands now, does not provide a definitive model to describe how players behave in trial-and-error processes. Your question is a good one, and we need more progress to come up with a clear-cut answer.

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Help Center

AA Abhinav Arya · 24 days ago (/learn/game-theory-introduction/discussions/D_OiV44kEeW7Cw7aHtmkKw/replies/cL_T_9vNEeW0gBLNziwTrw) ✓
But professor has explained that on the continuum of intelligence or rationality if the players are on extreme side of being highly intelligent and have required levels of reasoning (in location game with 50 points) he explained that such people are supposed to quickly converge to Nash equilibrium. So I think the answer should be yes. The problem now is only whether we accept that rationality implies logical reasoning. It is about the definitions of both. I think that they are somewhat the same. If I am rational I will choose a better outcome than some other worst outcome. So in this sense if I just know that the other person is also rational ...I will end up playing Nash equilibrium.

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 Sacha Schweizer · 24 days ago (/learn/game-theory-introduction/discussions/D_OiV44kEeW7Cw7aHtmkKw/replies/nv3Xe9vZEeWvQApnPujSmw) · Edited ✓
Real life situations are more complex than a 2x2 matrix. Even highly intelligent players would struggle to calculate a solution, and so they randomize. (In the traffic example, you try different roads according to the mood of the moment and eventually establish a personal statistic as a basis for more rational future decisions.) This is a good paradigm for finding a Nash equilibrium.

Sometimes the rational approach might even prove counterproductive: The staunch game theory ideologist sticks to what he perceives to be the better play, taking into account a temporary loss while hoping the other players converge to his solution, despite the broad public having already - more or less randomly - adopted another strategy. This would explain how users end up locked into suboptimal technologies despite the makers' struggle to social engineer them into the alleged optimal choice.

In the aggregate, the intelligence of the crowd that emerges compensates for the individuals' bounded rationality.

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SD (/learn/game-theory-introduction/profiles/f34069ce8df6de7dbebfedfb7e760d9f)

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